



Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: 3

Obtained Marks:

Introduction to Data Science

Assignment#04

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6. Report Requirements

1. INTRODUCTION

Air pollution has become one of the most serious environmental challenges worldwide. Rapid urbanization, industrial activities, and increasing vehicle emissions have significantly affected air quality in major cities. The Air Quality Index (AQI) is a widely used indicator that measures pollution levels and helps determine whether the air is safe or harmful for human health.

In this assignment, Reinforcement Learning (RL) is applied to develop a simple traffic control decision system based on AQI levels. A Q-Learning agent is trained to select appropriate traffic management actions according to different pollution conditions. The objective is to learn an optimal policy that minimizes pollution risks while applying suitable control measures.

2. OBJECTIVE

The main objectives of this project are:

- To convert AQI values into Reinforcement Learning states.
- To design a reward-based traffic control environment.
- To implement the Q-Learning algorithm.
- To train an intelligent agent for decision-making.
- To analyze the learned policy and evaluate its effectiveness.

3. DATASET DESCRIPTION

The project uses the Global Urban Air Quality Index Dataset (2015–2025).

Dataset Information:

Total Records: 3,660

Total Features: 13

Time Period: 2015–2025

Countries Covered: 9

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Target Variable: AQI

Generated RL States: Low AQI, Medium AQI, High AQI

The dataset contains pollutant measurements such as PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃, temperature, humidity, and wind speed.

4. RL ENVIRONMENT DESIGN

The Reinforcement Learning environment was designed using AQI values.

State Definition:

State 0 = Low AQI (≤ 100)

State 1 = Medium AQI (101–200)

State 2 = High AQI (> 200)

Action Definition:

Action 0 = No Restriction

Action 1 = Partial Restriction

Action 2 = High Pollution Alert

These actions represent possible traffic control measures under different pollution conditions.

5. AQI STATE DISTRIBUTION

The AQI values were categorized into three pollution states.

AQI State Distribution Table

Low AQI (≤ 100): 970 Records

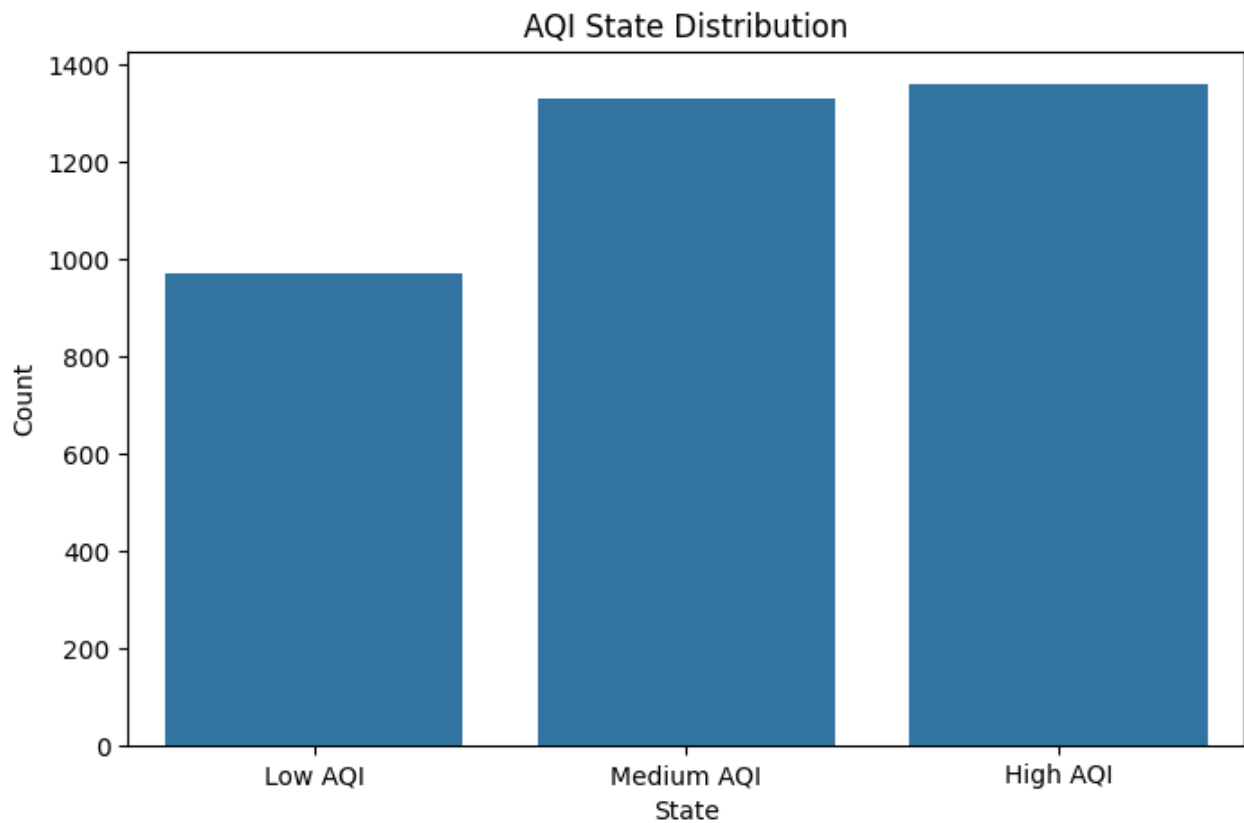
Medium AQI (101–200): 1,330 Records

High AQI (> 200): 1,360 Records

Analysis:

The dataset contains a balanced distribution of medium and high pollution conditions. High AQI states slightly dominate the dataset, indicating that polluted conditions occur frequently within the

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selected cities. This balanced distribution helps the Q-Learning agent experience different environmental situations during training.



6. REWARD STRUCTURE

The reward system was designed to encourage appropriate actions for each AQI level.

Reward Matrix

Low AQI:

No Restriction = +10

Partial Restriction = -2

High Pollution Alert = -5

Medium AQI:

No Restriction = -6

Partial Restriction = +10

High Pollution Alert = +2

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High AQI:

No Restriction = -10

Partial Restriction = -3

High Pollution Alert = +10

The reward values encourage the agent to select the most suitable action for each pollution level.

7. Q-LEARNING ALGORITHM

Q-Learning is a model-free Reinforcement Learning algorithm that learns the best action by maximizing cumulative rewards.

Training Parameters:

Learning Rate (Alpha): 0.1

Discount Factor (Gamma): 0.9

Exploration Rate (Epsilon): 0.2

Total Episodes: 500

The agent continuously interacts with the environment and updates the Q-table using the Bellman equation.

8. REWARD PER EPISODE ANALYSIS

The reward curve demonstrates the learning progress of the agent throughout training.

Training Statistics:

Total Episodes: 500

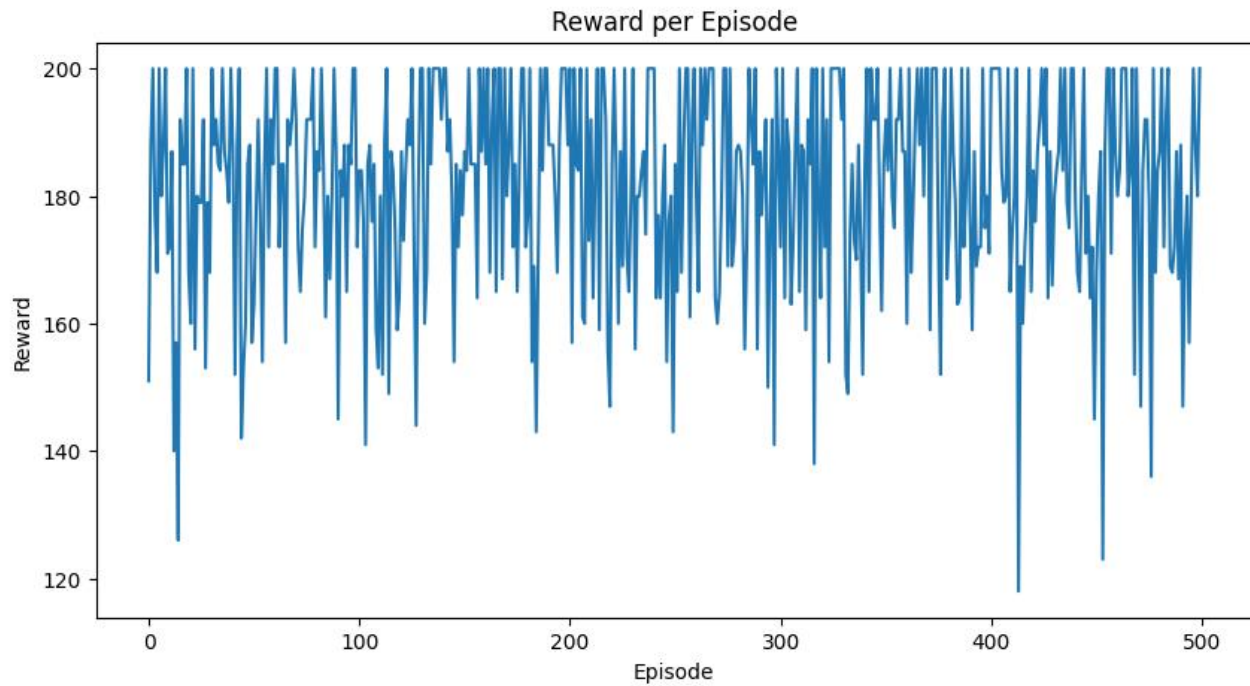
Maximum Reward Ceiling: 200

Exploration Strategy: Epsilon-Greedy

Epsilon Value: 0.2

Observation:

The reward values rapidly approached the maximum reward ceiling, indicating that the agent learned an effective strategy early in the training process. Small fluctuations in the graph occurred due to the exploration mechanism, which occasionally tested alternative actions.



9. FINAL Q-TABLE RESULTS

After training, the following Q-table was obtained.

Q-Table

State 0 (Low AQI)

No Restriction = 100.0

Partial Restriction = 88.0

High Pollution Alert = 85.0

State 1 (Medium AQI)

No Restriction = 84.0

Partial Restriction = 100.0

High Pollution Alert = 92.0

State 2 (High AQI)

No Restriction = 80.0

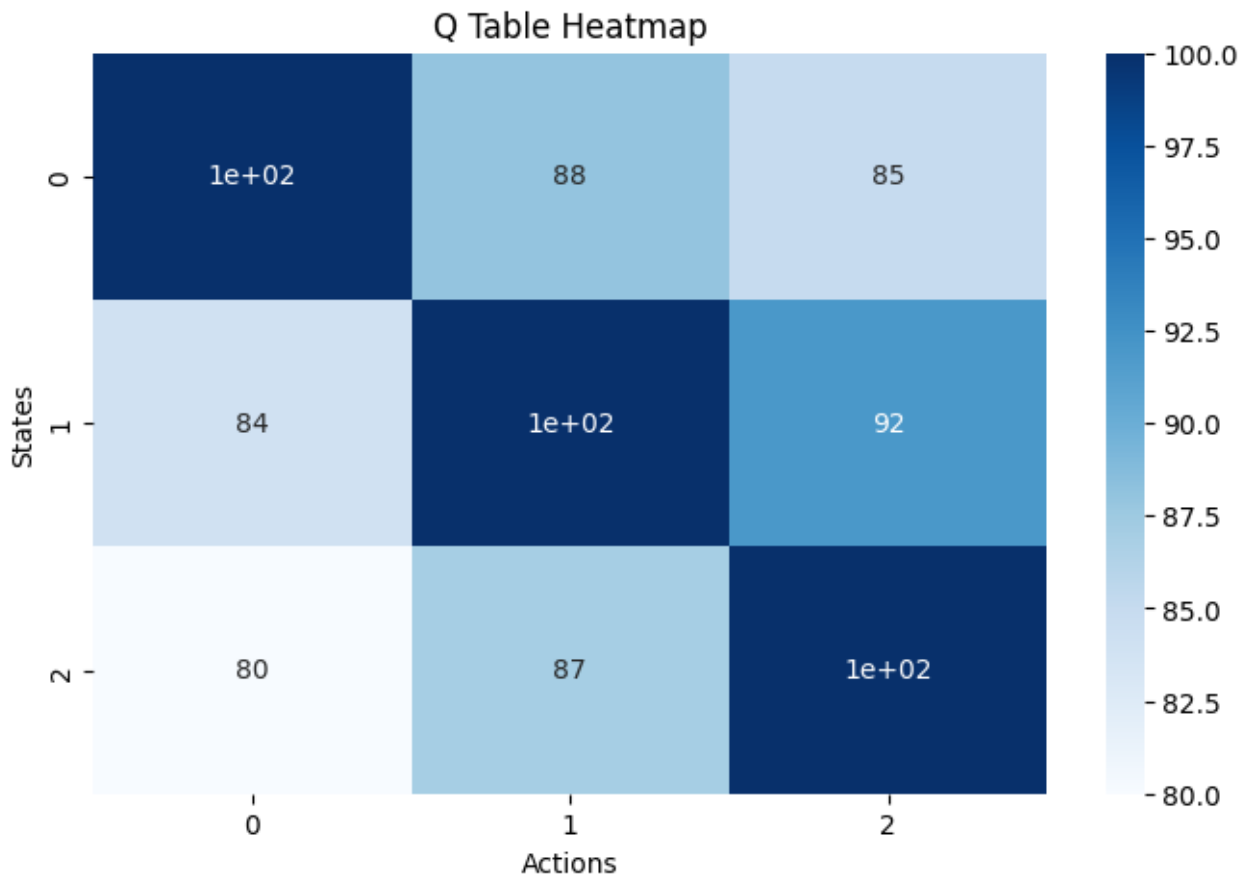
Partial Restriction = 87.0

High Pollution Alert = 100.0

Interpretation:

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The highest Q-value in each row corresponds to the optimal action. The results show that the agent successfully learned the best decision for every AQI state.



10. LEARNED POLICY

The final policy generated by the Q-Learning agent is shown below.

Policy Table

Low AQI → No Restriction

Medium AQI → Partial Restriction

High AQI → High Pollution Alert

The learned policy follows logical environmental rules and matches the intended reward design.

11. POLICY VALIDATION

To evaluate performance, the learned actions were compared with the expected optimal actions.

Expected Policy:

Low AQI → No Restriction

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Medium AQI → Partial Restriction

High AQI → High Pollution Alert

Learned Policy:

Low AQI → No Restriction

Medium AQI → Partial Restriction

High AQI → High Pollution Alert

Policy Alignment: 100%

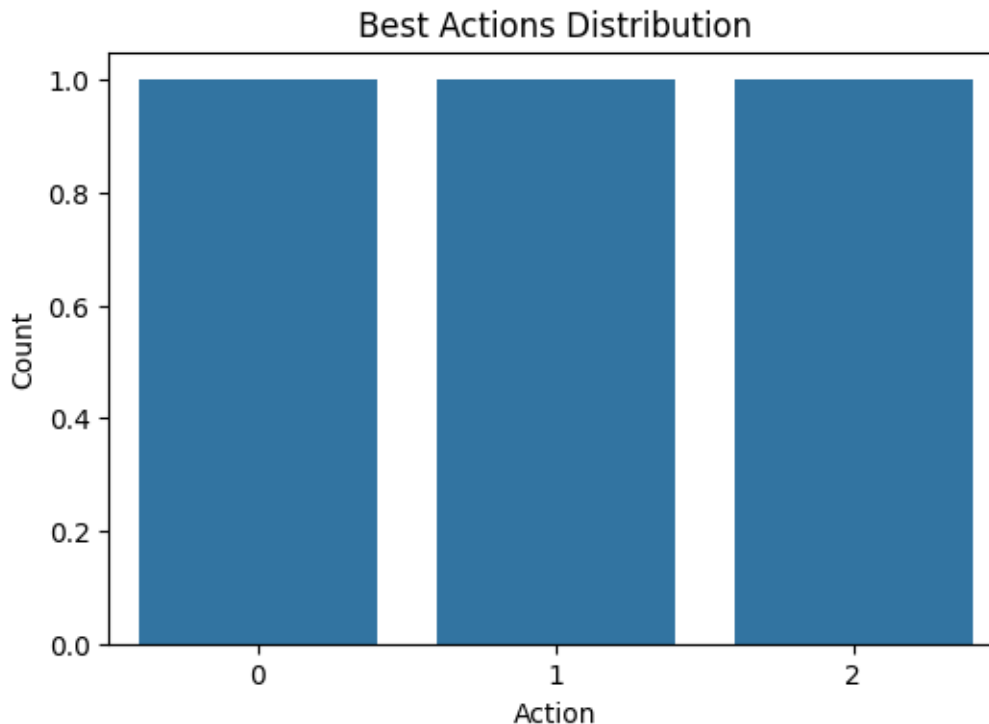
The agent successfully learned the optimal policy for all states.

12. ACTION DISTRIBUTION ANALYSIS

The action distribution graph illustrates the frequency of optimal actions selected by the trained agent.

Observation:

The agent consistently selected actions that correspond to the highest rewards. This confirms that the Q-Learning algorithm converged successfully and produced a stable policy.



13. DISCUSSION

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The implementation demonstrates how Reinforcement Learning can be applied to environmental decision-making systems. By converting AQI values into discrete states and assigning rewards to traffic control actions, the agent learned effective responses to different pollution conditions.

The results indicate that Q-Learning can be used as a simple decision-support technique for pollution management and smart city applications.

14. CONCLUSION

This assignment successfully applied Reinforcement Learning to an AQI-based traffic control problem. AQI values were categorized into three states, and a Q-Learning agent was trained using a reward-driven environment.

The final Q-table showed clear optimal actions for each state, while the reward curve demonstrated stable learning performance. The learned policy achieved 100% alignment with the expected policy, confirming that the agent effectively learned the intended traffic control strategy.

Future improvements may include larger state spaces, dynamic reward systems, and advanced Reinforcement Learning algorithms to simulate real-world environmental management scenarios.

15. REFERENCES

- Sutton, R. S., & Barto, A. G. Reinforcement Learning: An Introduction.
- Scikit-Learn Documentation.
- NumPy Documentation.
- Pandas Documentation.
- Global Urban Air Quality Index Dataset (2015–2025).

16. GITHUB REPOSITORY LINK

[<https://github.com/Qaim-Ali727/AQI-Data-Science-Project-/tree/main>]